

Himanshu Pandey

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github.com/Anzay12

SUMMARY

Data Science and ML Engineering student at IIT Madras with hands-on industry experience building end-to-end ML pipelines, deploying models on AWS, and engineering features for high-frequency financial data. Proficient in the full ML lifecycle from EDA and model development to MLOps, containerized deployment, and production monitoring. Seeking roles in Data Science or ML Engineering where I can drive measurable impact through scalable, production-grade machine learning systems.

EDUCATION

Indian Institute of Technology (IIT) Madras, Chennai, India

BS in Data Science and Applications

Sept. 2022 – Aug. 2026

Diploma in Programming & App Dev — Coursework: Data Structures, DBMS, Algorithms, Statistics

WORK EXPERIENCE

Shodhshala, Remote

Data Science & MLOps Intern

Jan. 2026 – Present.

- Designed and maintained **PostgreSQL** databases (1M+ rows of high-frequency financial data) with advanced query optimization and indexing strategies, reducing average query response time by ~30%.
- Built automated **ETL pipelines** in Python for ingesting and transforming large-scale trading datasets, cutting preprocessing time by ~40% and enabling real-time feature availability.
- Architected and deployed an **end-to-end MLOps pipeline** — spanning data ingestion, feature engineering, model training, experiment tracking (**MLflow**), and data versioning (**DVC + DagsHub**) — containerized via **Docker** and deployed on **AWS (EC2/S3)**, reducing model release cycles by ~50%.
- Configured **Grafana + Prometheus** monitoring dashboards for real-time model performance tracking, data drift detection, and automated alerting on latency and accuracy degradation in live trading strategies.
- Engineered **30+ features** from raw market data via modular Python scripts, improving model signal quality by ~15%; developed and backtested algorithmic trading strategies, improving average win rate by ~12%.

TECHNICAL PROJECTS

E-commerce RTO Prediction

Guide: github.com/Anzay12/Prediction_RTO Tech: Python, Gradient Boosting, Scikit-learn, Pandas

- Built a **Gradient Boosting classifier** achieving **68.75% F1-Score** and **73.33% ROC-AUC** to predict Return-to-Origin orders; benchmarked 5 models via cross-validation (CV: 70.66% ± 1.42%).
- Engineered 4 domain-specific features (Cost.Weight.Ratio, Weight.Bin, etc.) contributing **54%+ of total feature importance**; Weight and Discount identified as top RTO drivers.

Movie Audience Prediction — ML & Time Series

Guide: github.com/Anzay12/movie-audience-prediction-ml Tech: Python, XGBoost, LightGBM, Scikit-learn, Pandas

- Built an end-to-end forecasting pipeline with lag features (1/7/14/30-day) and rolling means across Random Forest, XGBoost, and LightGBM; selected best model via RMSE comparison.
- Engineered a weighted ensemble blending day-of-week patterns, 7 & 30-day trends, and theater-level behavior with multi-level fallback, improving prediction stability over single-model baseline.

SKILLS

Backend: Python, Flask, FastAPI, REST APIs, SQLAlchemy, Celery, Redis, JWT Authentication

Databases: PostgreSQL, SQLite, MongoDB, Query Optimization, Schema Design, ETL Pipelines

DevOps & Cloud: Docker, CI/CD, AWS (EC2, S3, IAM), Linux, Git

MLOps: MLflow, DVC, DagsHub, Grafana, Prometheus, Model Registry, Experiment Tracking, Data Versioning, Pipeline Automation

ML & Data: Scikit-learn, XGBoost, PyTorch, TensorFlow, Pandas, NumPy, EDA, Data Visualization

Frontend: React, Vue.js, JavaScript, HTML/CSS

CERTIFICATIONS

- Claude Code 101**
- MLOps Bootcamp: Mastering AI Operations**
- AWS Cloud Practitioner Essentials**
- Generative AI**

Anthropic
Udemy
GeeksforGeeks
Udemy